

Shangyi Zhou

Brooke Luetgert

DIGS 20004

26 May 2026

Statement of Collaboration

This project was completed individually. Claude was used as an AI assistant throughout the development process.

The overall structure of the dashboard was adapted from app.py of the Week 7 module as a template. I made all decisions regarding which dataset to use, which questions to ask of the data, and which visualizations to build.

Claude was consulted for specific technical problems I could not resolve on my own. These include: illustrating quarterly labels for the RangeSlider using `dt.to_period("Q")`; calculating per-year percentage share using `groupby` and `transform`; reshaping data into a matrix using `pivot()` for the heatmap; rendering the heatmap using `px.imshow()`; building the bubble map using `px.scatter_geo()` with manually defined regional coordinates; controlling the hover tooltip using `hover_data`; and resolving callback errors caused by tab-internal components using `suppress_callback_exceptions=True`; etc. The specified comments on AI use are also included in the py file.

The Write-Up, data selection, research questions, narrative text inside the dashboard, and all design decisions were my own. I would assess my contribution as full ownership of the project concept and structure, with AI assistance limited to specific implementation details that I could not figure out from the template alone.